Multi-wavelength Astronomy: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — A visible photon. Return →

Use $\nu = c/\lambda = 2.998 \times 10^{8}/(5.00 \times 10^{-7}) = 5.996 \times 10^{14}$ Hz. The energy is $E = h\nu = 3.97 \times 10^{-19}$ J. Since $hc = 1240$ eV nm, $E = 1240/500 = 2.48$ eV. **Check:** visible photons are a few eV, so the answer is physically reasonable.

Solution 2 — Two thermal peaks. Return →

Wien's law gives $T = 2.898 \times 10^{-3}/\lambda_{\max}$. For 500 nm, $T = 5796$ K, close to the Sun's effective temperature. For 10 μm, $T = 289.8$ K, close to room temperature. **Check:** hotter objects peak at shorter wavelengths.

Solution 3 — A shifted line. Return →

The redshift is $1 + z = 608/121.6 = 5$, so $z = 4$. Every rest wavelength is multiplied by 5, so 500 nm is observed at 2500 nm, in the infrared. **Check:** the same factor applies to all features from the same source.

Solution 4 — Magnitude contrast. Return →

The magnitude relation is $m_1 - m_2 = -2.5 \log_{10}(F_1/F_2)$. A difference of 5 mag corresponds to a flux ratio of $10^{0.4 \times 5} = 100$. The smaller magnitude is brighter. **Check:** five magnitudes is exactly the historical factor of 100.

Solution 5 — Ground or space? Return →

21 cm radio is ground-accessible because the atmosphere is transparent at many radio wavelengths. 550 nm optical is ground-accessible through the optical window. 200 nm ultraviolet is strongly absorbed by ozone and generally requires space. 1 keV X-rays are absorbed by the atmosphere and require space. **Check:** the decision depends on atmospheric transmission, not on whether the source is bright.

Solution 6 — Dust penalty. Return →

Flux is reduced by $10^{0.4 A_V} = 10^{0.8} = 6.31$. Only about 16% of the visual flux remains. Dust extinction is usually weaker at longer wavelengths, so infrared observations can penetrate dusty regions more effectively. **Check:** larger extinction in magnitudes means a larger flux reduction.

Solution 7 — Counting X-rays. Return →

One 1 keV photon has energy $1000 \times 1.602 \times 10^{-19} = 1.602 \times 10^{-16}$ J. The rate is $10^{-15}/1.602 \times 10^{-16} = 6.24$ photons/s. X-ray sources are often faint enough that individual photon events are counted, and each photon's energy can be recorded. **Check:** a few photons per second is a plausible faint X-ray count rate.

Solution 8 — Radio versus optical resolution. Return →

Use $\theta = 1.22\lambda/D$ in radians and multiply by 206265. For the radio dish, $\theta = 1.22(0.21)/25 = 0.01025$ rad $= 2110''$, about 0.59°. For the optical telescope, $\theta = 1.22(5.0 \times 10^{-7})/0.10 = 6.10 \times 10^{-6}$ rad $= 1.26''$. **Check:** long wavelengths need much larger apertures or interferometry for fine resolution.

Solution 9 — A high-redshift thermal peak. Return →

The rest-frame peak is $2.898 \times 10^{-3}/10^4 = 2.898 \times 10^{-7}$ m, or 290 nm, in the ultraviolet. At $z = 2$, the observed peak is $3 \times 290 = 870$ nm, in the near-infrared. **Check:** hot stars can have their rest-frame UV light shifted into the infrared by cosmological expansion.

Solution 10 — Design a campaign. Return →

Infrared imaging and spectroscopy constrain dust emission, obscured star formation and redshifted stellar or nebular features. Radio continuum and line observations constrain synchrotron emission, neutral or molecular gas, and star-formation tracers less affected by dust. Optical or near-infrared spectroscopy can still provide line ratios and kinematics where extinction permits. X-rays could test for an active nucleus or hot gas. A key issue is matching angular resolution: a large radio beam may blend regions that are separate in the infrared, so photometry must use comparable apertures or beam-matched maps.